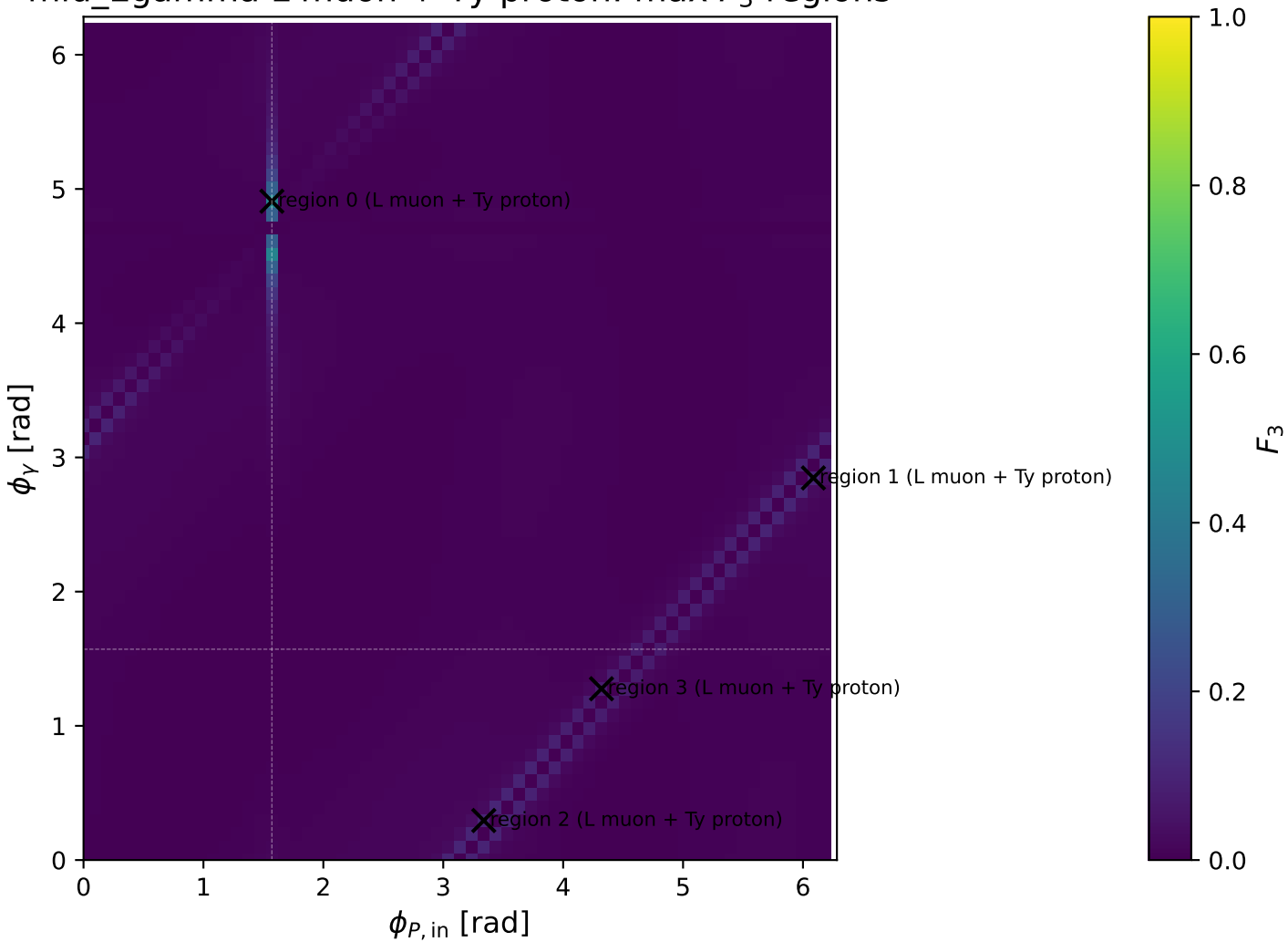
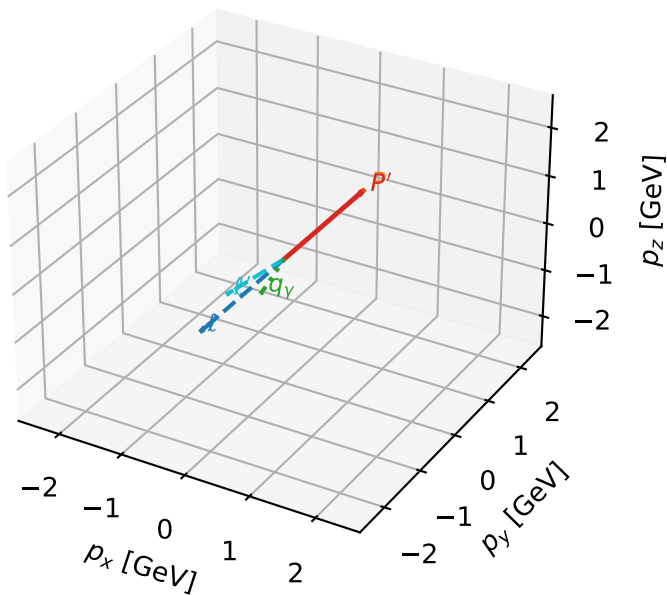


mid_Egamma L muon + Ty proton: max F_3 regions



L muon + Ty proton characteristic max F_3 configuration

3D momenta

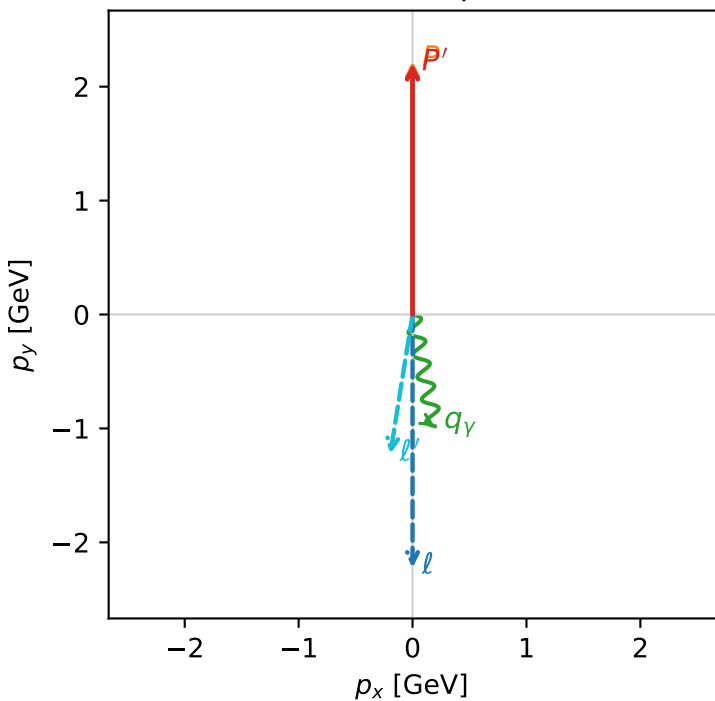


f3_mid_egamma_lty_region_0 $F_3=0.445324$
 region: mid_Egamma_LTy_region_0
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|L_\mu\rangle \otimes |T_{Y,p}\rangle$

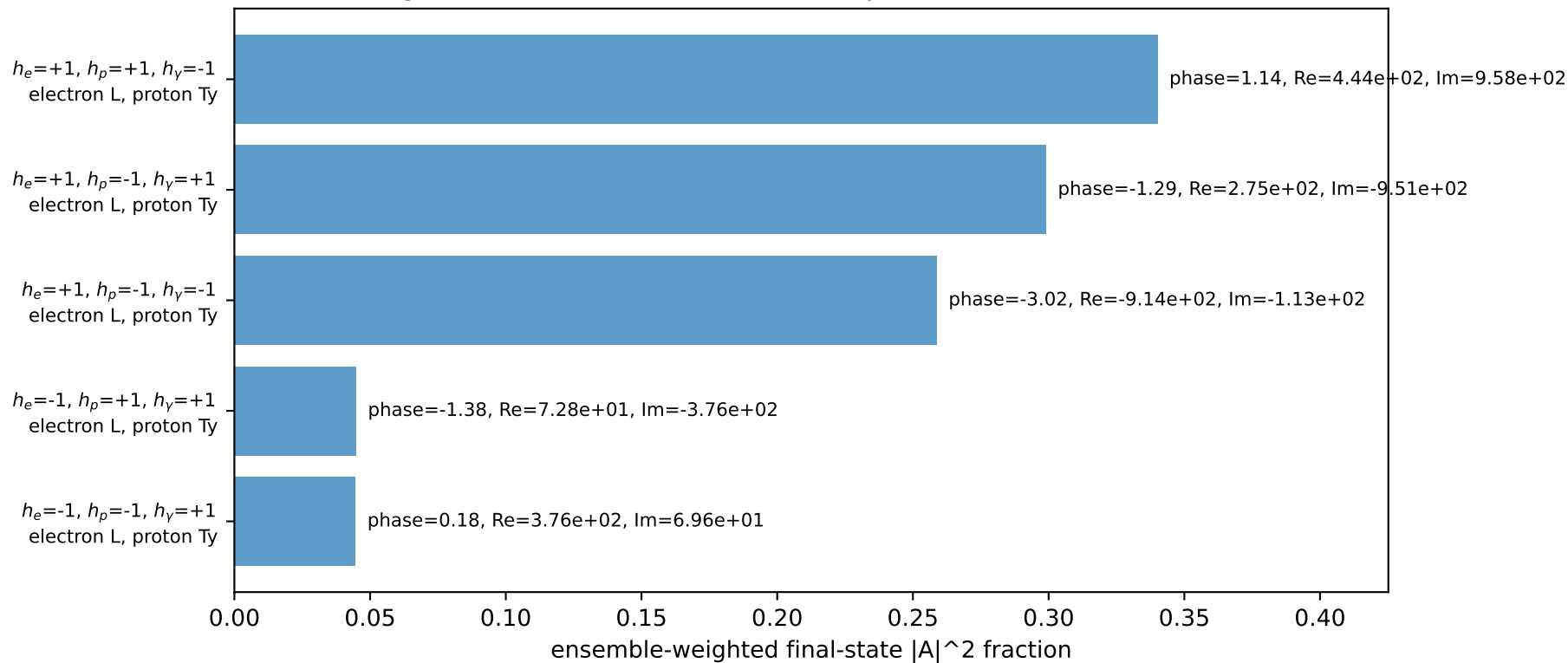
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=2.20402$
 $\theta_{in}=1.57079$, $\phi_\ell=4.71239$, $\phi_P=1.5708$
 $E_Y=1$, $\phi_Y=4.90874$
 $Q^2=0.0726476$, $x_B=0.00790796$, $t=-5.54648e-05$, $W^2=9.99383$, $y=0.445416$
 $\theta(\ell', \gamma)=20.3116$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2256$ $p_x= -0.0000$ $p_y= -2.2231$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= 0.0000$ $p_y= 2.2231$ $p_z= 0.0000$
 ℓ' $E= 1.2432$ $p_x= -0.1951$ $p_y= -1.2232$ $p_z= 0.0000$
 P' $E= 2.3953$ $p_x= 0.0000$ $p_y= 2.2040$ $p_z= 0.0000$
 q_Y $E= 1.0000$ $p_x= 0.1951$ $p_y= -0.9808$ $p_z= 0.0000$

Transverse plane

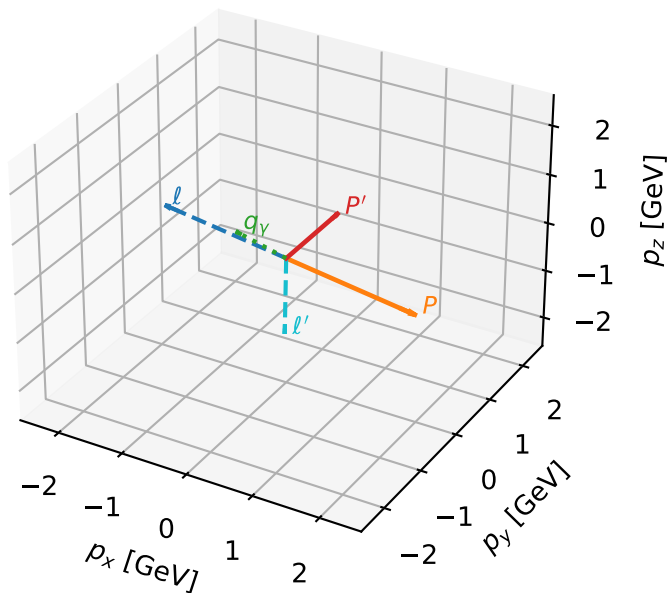


Leading initial-ensemble/final-state components (N=5, retained=98.8%)



L muon + Ty proton characteristic max F_3 configuration

3D momenta



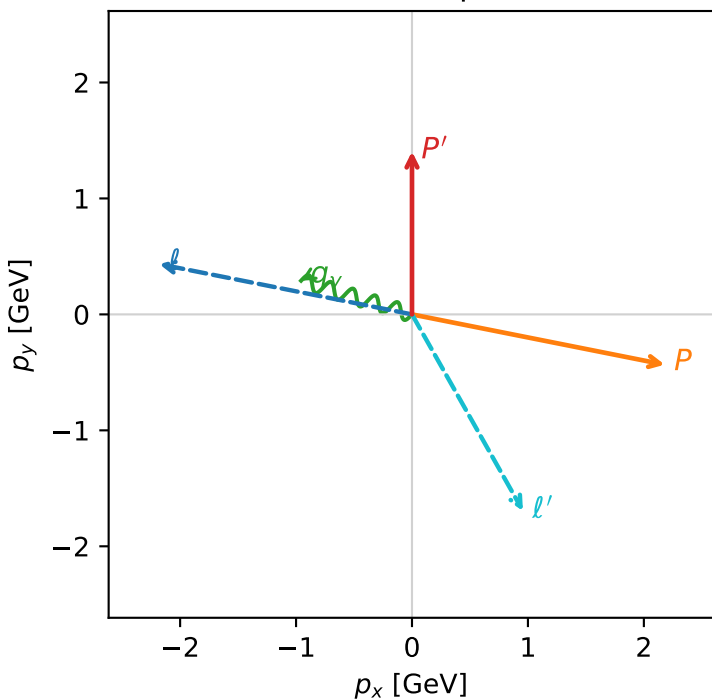
f3_mid_egamma_lty_region_1 $F_3=0.103977$
 region: mid_Egamma_LTy_region_1
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|L_\mu\rangle \otimes |T_{y,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=1.40476$
 $\theta_{in}=1.57079$, $\phi_t=2.94524$, $\phi_P=6.08684$
 $E_\gamma=1$, $\phi_\gamma=2.84707$
 $Q^2=14.2981$, $x_B=0.848008$, $t=-7.61026$, $W^2=3.44256$, $y=0.817503$
 $\theta(l', \gamma)=136.322$ deg

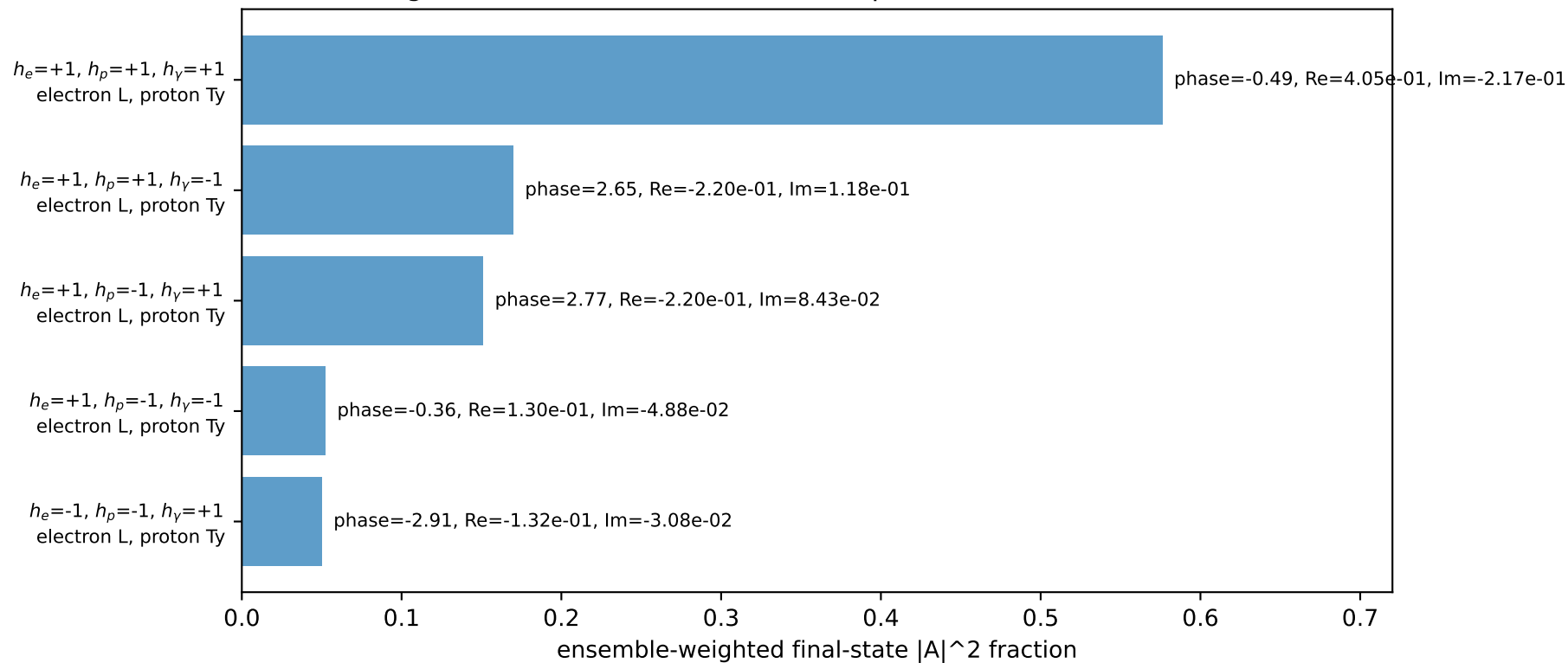
four-momenta $[E, p_x, p_y, p_z]$ GeV:

l $E= 2.2256$ $p_x= -2.1804$ $p_y= 0.4337$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= 2.1804$ $p_y= -0.4337$ $p_z= 0.0000$
 l' $E= 1.9494$ $p_x= 0.9569$ $p_y= -1.6950$ $p_z= 0.0000$
 P' $E= 1.6891$ $p_x= 0.0000$ $p_y= 1.4048$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= -0.9569$ $p_y= 0.2903$ $p_z= 0.0000$

Transverse plane

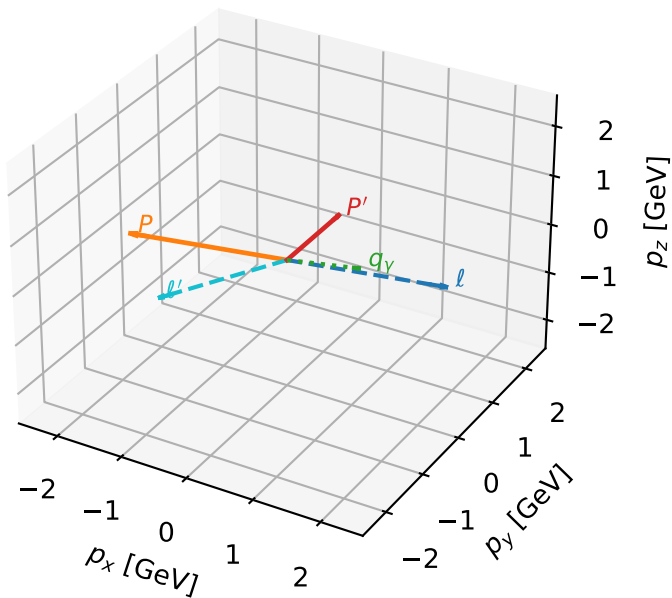


Leading initial-ensemble/final-state components (N=5, retained=99.9%)



L muon + Ty proton characteristic max F_3 configuration

3D momenta

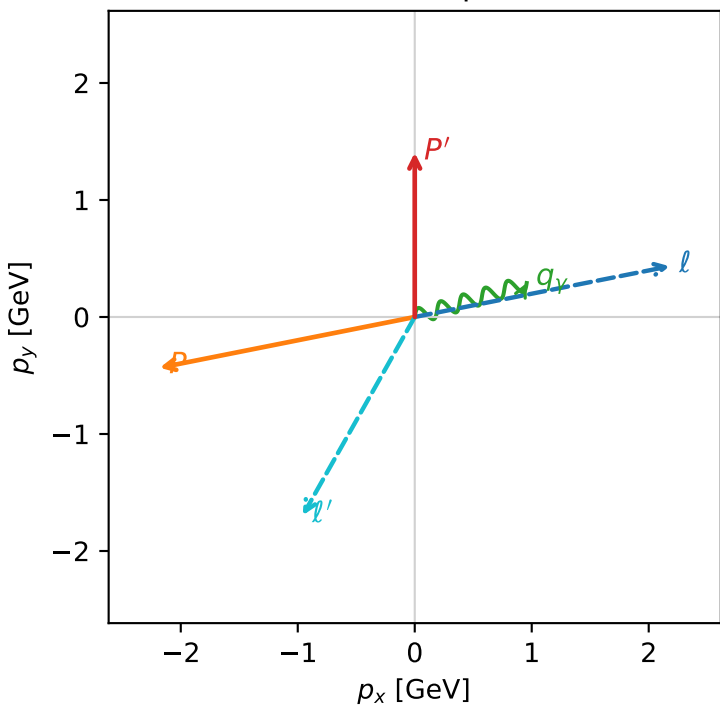


f3_mid_egamma_lty_region_2 F_3=0.103977
 region: mid_Egamma_LTy_region_2
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|L_\mu\rangle \otimes |T_{Y,p}\rangle$

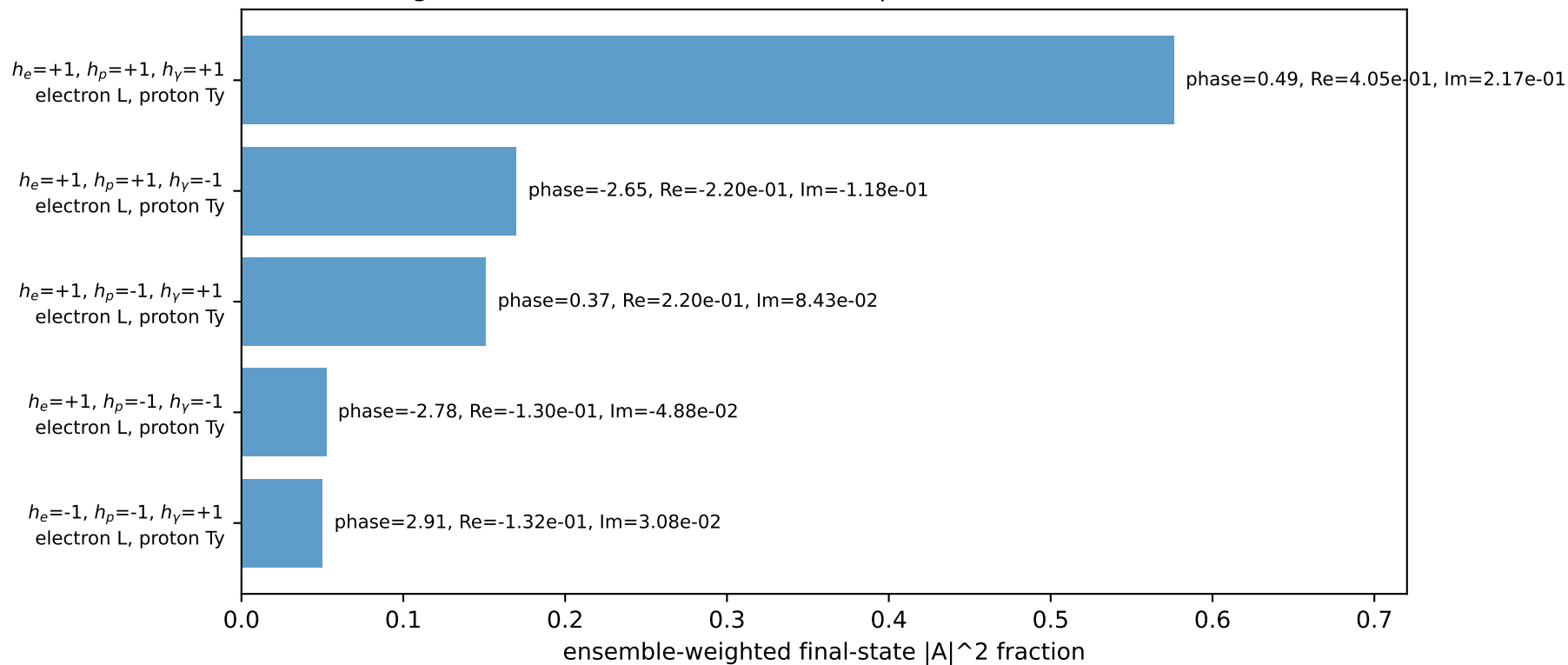
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=1.40476$
 $\theta_{in}=1.57079$, $\phi_l=0.19635$, $\phi_p=3.33794$
 $E_Y=1$, $\phi_Y=0.294524$
 $Q^2=14.2981$, $x_B=0.848008$, $t=-7.61026$, $W^2=3.44256$, $y=0.817503$
 $\theta(l', \gamma)=136.322$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= 2.1804$ $p_y= 0.4337$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= -2.1804$ $p_y= -0.4337$ $p_z= 0.0000$
 l' $E= 1.9494$ $p_x= -0.9569$ $p_y= -1.6950$ $p_z= 0.0000$
 P' $E= 1.6891$ $p_x= 0.0000$ $p_y= 1.4048$ $p_z= 0.0000$
 q_Y $E= 1.0000$ $p_x= 0.9569$ $p_y= 0.2903$ $p_z= 0.0000$

Transverse plane

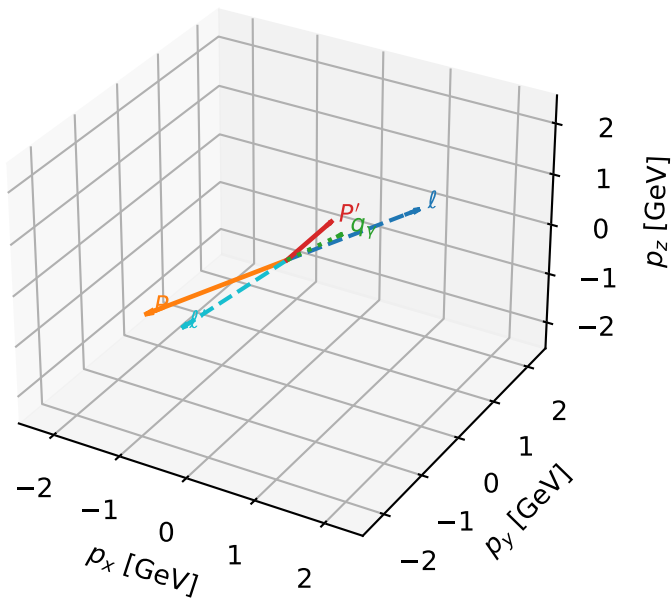


Leading initial-ensemble/final-state components (N=5, retained=99.9%)



L muon + Ty proton characteristic max F_3 configuration

3D momenta

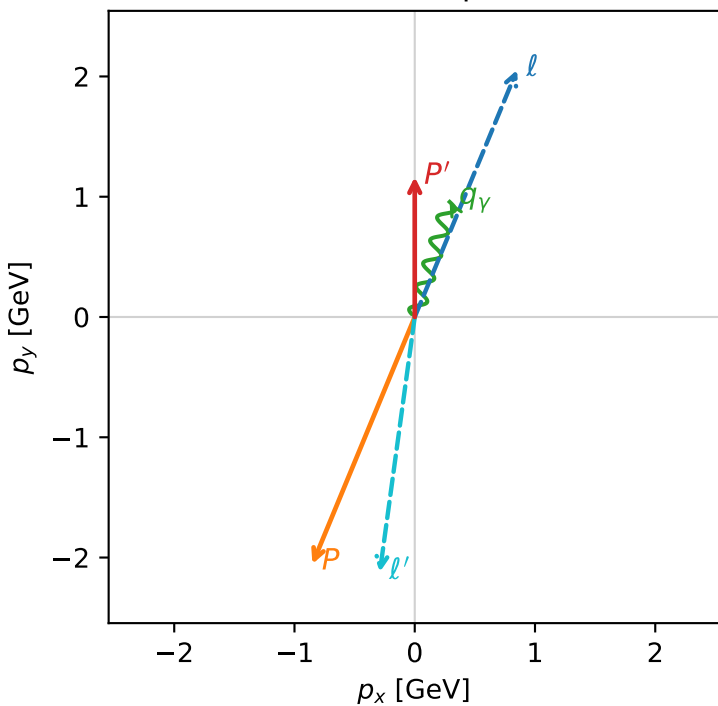


f3_mid_egamma_lty_region_3 F_3=0.0768473
 region: mid_Egamma_LTy_region_3
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|L_\mu\rangle \otimes |T_{\gamma,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=1.16417$
 $\theta_{in}=1.57079$, $\phi_l=1.1781$, $\phi_p=4.31969$
 $E_\gamma=1$, $\phi_\gamma=1.27627$
 $Q^2=18.7258$, $x_B=0.960899$, $t=-10.2372$, $W^2=1.64184$, $y=0.944869$
 $\theta(l', \gamma)=170.918$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= 0.8507$ $p_y= 2.0539$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= -0.8507$ $p_y= -2.0539$ $p_z= 0.0000$
 l' $E= 2.1435$ $p_x= -0.2903$ $p_y= -2.1211$ $p_z= 0.0000$
 P' $E= 1.4950$ $p_x= 0.0000$ $p_y= 1.1642$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= 0.2903$ $p_y= 0.9569$ $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=3, retained=99.4%)

